



# Complete Technical Report

All changes, infrastructure, and features built — Dec 25, 2025 to present

|                                 |                                    |
|---------------------------------|------------------------------------|
| <b>Report Date:</b>             | March 14, 2026                     |
| <b>Period:</b>                  | December 25, 2025 — March 13, 2026 |
| <b>Total Database Tables:</b>   | 71                                 |
| <b>Total Decision Receipts:</b> | 66,220                             |
| <b>Total Evaluation Cycles:</b> | 750,100                            |
| <b>Classification:</b>          | Internal — Not for distribution    |

## Table of Contents

1. Executive Summary
2. Timeline — Dec 2025 to Jan 2026
3. Timeline — Feb 2026 (First Half)
4. Timeline — Feb 2026 (Second Half)
5. Timeline — Mar 2026 (First Week)
6. Timeline — Mar 2026 (Second Week)
7. Database — B2B Client Tables
8. Database — Sandbox Tables
9. Database — Alert System Tables
10. Database — Governance Receipts (PQC-signed)
11. Database — 5 Compliance Modules
12. Database — Decision Engine & Vetos
13. Database — Backtesting Phase 0
14. Database — Trading (Real & Paper)
15. Database — Users & Configuration
16. API Endpoints — B2B External
17. API Endpoints — Sandbox
18. API Endpoints — Alerts
19. API Endpoints — Compliance Modules

- 20. Web Pages ([omnixquantum.net](https://omnixquantum.net))
- 21. Technical Validation — Terra/LUNA Forensic Simulation (May 2022)
- 22. Documents Generated
- 23. Summary in Numbers

# 1. Executive Summary

This report documents every significant change, feature, and infrastructure component built into OMNIX from December 25, 2025 through March 13, 2026. It covers database tables (71 total), API endpoints, web pages, compliance modules, and generated documents. All database row counts are queried live from the local PostgreSQL instance at generation time.

Key numbers: 750,100 evaluation cycles processed, 66,220 PQC-signed receipts generated, 25,681 vetos logged, 2,237 real Kraken trades recorded, 5 B2B clients registered, 4 vertical demos live.

## 2. Timeline — Dec 2025 to Jan 2026

### Foundation of the public website and bot stabilization

- Created the public website (omnixquantum.net) with separate commercial and institutional pages
- Added contact sections with WhatsApp, FAQ, competitor comparison table
- Connected frontend with Railway backend correctly
- Improved bot AI response speed (timeout optimization and model selection)
- Upgraded primary AI model to Gemini 2.5 Flash
- Configured dynamic Track Record day counter (no more hardcoded number)
- Implemented dashboard caching for faster loading
- Fixed navigation and unknown route redirects
- Added leadership team section to institutional page
- Configured Railway deployment with Python + Node.js build pipeline

## 3. Timeline — Feb 2026 (First Half)

### Repositioning as Decision Governance + Multi-vertical demos

- OMNIX officially repositioned as 'Decision Governance Infrastructure' — not a trading bot
- Added voice interface for hands-free governance interaction
- Implemented AI response compression (contextual: institutional vs casual tone)
- Built Insurance Governance Demo — underwriting with governance checkpoints
- Built Credit/Lending Governance Demo — credit risk evaluation
- Built Energy Trading Governance Demo — natural gas, crude oil, solar, wind, LNG, electricity
- Built Biotech/Clinical Trial Governance Demo — real ClinicalTrials.gov NCT data
- Updated bot tone to be more institutional, less servile
- Improved AI greeting detection and natural language handling
- Created mentor packages with personalized draft messages and resources

## 4. Timeline — Feb 2026 (Second Half)

### B2B infrastructure, security, and investor documents

- Built B2B External Governance API with API key authentication (X-API-Key header)
- Implemented RBAC (Role-Based Access Control) — regular clients vs admin with separate permissions
- Added payload encryption at rest using Fernet (AES-128-CBC + HMAC-SHA256)
- Built 5 Compliance Modules aligned with NIST AI RMF, ISO/IEC 42001, and EU AI Act
- Module 1: Risk Mapping — use case classification, impact scoring, stakeholder mapping
- Module 2: Measurement & Monitoring — checkpoint metrics, drift detection
- Module 3: Human Oversight — override logging with PQC-signed audit trail
- Module 4: Incident Management — incident reporting, review, and corrective actions

- Module 5: Governance Reporting — period-based reports with content hashing
- Generated research paper PDF with reproducibility section
- Created 3-tier security documentation: public, institutional, and internal
- Removed Ivan Guzman from all files — Harold Nunes as sole visible Founder
- Launched public deployment on Railway + custom domain omnixquantum.net
- Generated multiple investor PDF documents and mentor packages
- Clarified PQC communication rules (no FIPS 204, use 'NIST-standardized')
- Implemented API key rotation and database backup documentation

## 5. Timeline — Mar 2026 (First Week)

### Eureka Dubai preparation and presentation materials

- Created complete pitch deck with professional slides for Eureka Dubai (March 18)
- Generated Business Model Canvas PDF
- Created executive report for investors
- Prepared presentation script with real operational data
- Built Q&A; simulation guide (tough investor questions with prepared answers)
- Added Brazil financial regulations context for channel partner Richard (Blue Mesh)
- Updated AI Risk Analyst report with independent validation
- Fixed Black Swan detection classification logic
- Made all website and contact links clickable in generated reports

## 6. Timeline — Mar 2026 (Second Week)

### 8-checkpoint architecture + real-time data pipeline

- Expanded governance pipeline from 6 to 8 checkpoints: added CP-0 (Signal Integrity) and CP-7 (Temporal Coherence)
- Built interactive Sandbox for B2B clients — test evaluations without API key
- Built configurable Alert System — Slack, email, webhook notifications per client
- Added per-client threshold customization — each company adjusts their own checkpoint thresholds
- Fixed all pages to consistently show 8 checkpoints (Energy, Biotech, Credit, Insurance demos)
- Connected Railway public API to return all live metric fields (decisions\_blocked, capital\_preserved\_pct, system\_uptime\_days, verticals\_demo)
- Made system\_uptime\_days calculate automatically each day from Jan 15, 2026
- Eliminated all remaining hardcoded values in the web (22,000+, 98.5%, -1.5%)
- All web pages now refresh metrics every 60 seconds from Railway production API

## 7. Database — B2B Client Tables

| Table             | Rows | Columns (from DB schema)                                                            |
|-------------------|------|-------------------------------------------------------------------------------------|
| b2b_clients       | 5    | id, client_id, api_key_hash, name, email, role, is_active, created_at, last_seen_at |
| client_thresholds | 0    | id, client_id, checkpoint_id, threshold, updated_by, updated_at, created_at         |

b2b\_clients stores registered API clients with hashed keys and RBAC roles. client\_thresholds allows per-client checkpoint threshold customization with audit trail.

## 8. Database — Sandbox Tables

| Table               | Rows | Columns (from DB schema)                                                                          |
|---------------------|------|---------------------------------------------------------------------------------------------------|
| sandbox_sessions    | 3    | id, session_id, client_id, name, description, evaluation_count, created_at, expires_at, last_eval |
| sandbox_evaluations | 6    | id, evaluation_id, session_id, client_id, asset, domain, signals, decision, decision_score, check |

The sandbox allows potential clients to test the governance engine without authentication. Sessions expire automatically. Each evaluation generates a full 8-checkpoint trace.

## 9. Database — Alert System Tables

| Table         | Rows | Columns (from DB schema)                                                                              |
|---------------|------|-------------------------------------------------------------------------------------------------------|
| alert_configs | 2    | id, alert_config_id, client_id, alert_type, channel, enabled, config, created_at, updated_at          |
| alert_events  | 1    | id, event_id, alert_config_id, client_id, alert_type, channel, trigger_data, delivery_status, deliver |

## 10. Database — Governance Receipts (PQC-signed)

| Table                    | Rows   | Columns (from DB schema)                                                                          |
|--------------------------|--------|---------------------------------------------------------------------------------------------------|
| decision_receipts        | 66,220 | id, receipt_id, timestamp_utc, asset, decision, veto_chain, policy_version, engine_version, prev  |
| exit_governance_receipts | 78     | id, receipt_id, position_id, symbol, exit_reason, regime, should_exit, gate1_threshold_verdict, g |

Every governance decision generates a cryptographically signed receipt using Dilithium-3 (NIST-standardized post-quantum algorithm). Receipts form a SHA-256 hash chain for tamper detection. Exit governance receipts track the 3-gate exit pipeline with regime-adjusted thresholds.

## 11. Database — 5 Compliance Modules (NIST AI RMF / EU AI Act)

| Table                       | Rows | Module — Columns (from DB schema)                                                                                                |
|-----------------------------|------|----------------------------------------------------------------------------------------------------------------------------------|
| governance_risk_map         | 0    | Module 1 — Risk Mapping: id, client_id, use_case, classification, impact_financial, impact_operational, risk_score, last_updated |
| governance_metrics          | 0    | Module 2 — Measurement: id, client_id, checkpoint_id, approval_rate, block_rate, avg_score, v                                    |
| governance_drift_log        | 0    | Module 2 — Drift Detection: id, client_id, signal_name, baseline_stats, current_stats, drift_score                               |
| governance_overrides        | 0    | Module 3 — Human Oversight: id, client_id, receipt_id, decision_before, decision_after, justification                            |
| governance_incidents        | 0    | Module 4 — Incidents: id, client_id, incident_id, severity, title, description, related_receipt_id, status                       |
| governance_incident_reviews | 0    | Module 4 — Reviews: id, incident_id, reviewer, findings, corrective_actions, reviewed_at, created_at                             |
| governance_reports          | 0    | Module 5 — Reporting: id, client_id, period_start, period_end, report_type, content_json, content_md5                            |

Five additive governance modules aligned with NIST AI RMF, ISO/IEC 42001, and the EU AI Act. Each module introduces dedicated PostgreSQL tables and REST endpoints. All override operations generate PQC-signed audit trails.

## 12. Database — Decision Engine & Vetos

| Table                      | Rows    | Columns (from DB schema)                                                                        |
|----------------------------|---------|-------------------------------------------------------------------------------------------------|
| shadow_trade_events        | 750,100 | id, created_at, symbol, market, intended_action, intended_quantity, intended_entry_price, inter |
| trading_veto_log           | 25,681  | id, created_at, veto_type, engine_stage, symbol, market, user_id, trade_id, blocked_capital, cu |
| filter_calibration_metrics | 0       | id, calculated_at, period_start, period_end, veto_type, symbol, market, total_vetoes, vetoes_wi |
| shadow_trade_outcomes      | 50      | id, shadow_event_id, calculated_at, price_at_veto, price_after_1h, price_after_4h, price_after  |

## 13. Database — Backtesting Phase 0

| Table                   | Rows | Columns (from DB schema)                                                                       |
|-------------------------|------|------------------------------------------------------------------------------------------------|
| backtest_phase0_results | 225  | id, eval_key, asset, eval_hour, decision, veto_chain, checkpoints_passed, actual_pnl_usd, cou  |
| backtest_phase0_signals | 258  | id, eval_key, asset, eval_hour, probability_score, risk_exposure, signal_coherence, trend_pers |
| backtest_phase0_ohlcv   | 0    | id, asset, candle_time, open, high, low, close, volume, fetched_at                             |

## 14. Database — Trading (Real & Paper)

| Table                       | Rows  | Columns (from DB schema)                                                                                |
|-----------------------------|-------|---------------------------------------------------------------------------------------------------------|
| kraken_real_trades          | 2,237 | id, txid, refid, trade_time, type, subtype, aclass, asset, amount, fee, balance, amount_usd, phase      |
| paper_trading_trades        | 119   | id, user_id, symbol, side, quantity, entry_price, exit_price, profit_loss, profit_pct, strategy, status |
| paper_trading_balances      | 1     | id, user_id, balance_usd, btc_balance, eth_balance, total_trades, winning_trades, losing_trades         |
| paper_trading_daily_reports | 1     | id, user_id, report_date, balance_usd, daily_pnl, period_pnl, roi_pct, win_rate_pct, sharpe_ratio       |



| Table  | Rows | Columns (from DB schema)                                                                        |
|--------|------|-------------------------------------------------------------------------------------------------|
| trades | 0    | id, user_id, symbol, side, quantity, price, total, fee, order_id, exchange, status, executed_at |

## 15. Database — Users & Configuration

| Table         | Rows | Columns (from DB schema)                                                                            |
|---------------|------|-----------------------------------------------------------------------------------------------------|
| users         | 14   | user_id, telegram_id, username, first_name, last_name, language_code, is_premium, is_admin          |
| user_settings | 4    | user_id, username, risk_profile, trading_limits, protection_settings, notification_preferences, all |
| conversations | 110  | id, user_id, user_message, ai_response, language, created_at                                        |
| system_config | 0    | id, config_key, config_value, description, updated_at                                               |

## 16. API Endpoints — B2B External (authenticated with X-API-Key)

| Method | Endpoint                                      | Function                                               |
|--------|-----------------------------------------------|--------------------------------------------------------|
| POST   | /api/governance/evaluate                      | Run 8-checkpoint evaluation, return PQC-signed receipt |
| GET    | /api/governance/receipts                      | List client receipts (isolated by client_id)           |
| GET    | /api/governance/schema                        | Documentation of accepted signals                      |
| POST   | /api/governance/admin/clients                 | Create client and generate API key (admin)             |
| DELETE | /api/governance/admin/clients/<id>            | Deactivate client (admin)                              |
| POST   | /api/governance/admin/clients/<id>/rotate     | Rotate API key (admin)                                 |
| GET    | /api/governance/admin/clients/<id>/thresholds | Get client thresholds (admin)                          |
| PUT    | /api/governance/admin/clients/<id>/thresholds | Set per-client thresholds (admin)                      |
| DELETE | /api/governance/admin/clients/<id>/thresholds | Revert to defaults (admin)                             |
| GET    | /api/governance/metrics                       | Public real-time governance metrics                    |
| GET    | /api/verify/<receipt_id>                      | Public receipt verification                            |

## 17. API Endpoints — Sandbox (no authentication)

| Method | Endpoint                              | Function                       |
|--------|---------------------------------------|--------------------------------|
| POST   | /api/governance/sandbox/sessions      | Create test session            |
| GET    | /api/governance/sandbox/sessions      | List sessions                  |
| GET    | /api/governance/sandbox/sessions/<id> | Get session details            |
| DELETE | /api/governance/sandbox/sessions/<id> | Delete session                 |
| POST   | /api/governance/sandbox/evaluate      | Run evaluation without API key |
| GET    | /api/governance/sandbox/schema        | Signal schema documentation    |

## 18. API Endpoints — Alerts

| Method | Endpoint                           | Function                   |
|--------|------------------------------------|----------------------------|
| PUT    | /api/governance/alerts/config      | Create alert configuration |
| GET    | /api/governance/alerts/config      | List client alert configs  |
| POST   | /api/governance/alerts/test        | Test an alert              |
| DELETE | /api/governance/alerts/config/<id> | Delete alert config        |

## 19. API Endpoints — Compliance Modules

| Module           | Endpoints                                  | Function                                  |
|------------------|--------------------------------------------|-------------------------------------------|
| Risk Mapping     | POST/GET /api/governance/risk-map          | Create and list risk maps per client      |
| Monitoring       | GET /api/governance/metrics/live           | Live approval/block statistics            |
| Drift Detection  | POST /api/governance/drift/detect          | Detect signal drift vs baseline           |
| Human Oversight  | POST /api/governance/overrides             | Log decision overrides with PQC signature |
| Incidents        | POST/GET /api/governance/incidents         | Report and track governance incidents     |
| Incident Reviews | POST /api/governance/incidents/<id>/review | Submit incident review findings           |
| Reporting        | POST/GET /api/governance/reports           | Generate and retrieve governance reports  |

## 20. Web Pages (omnixquantum.net)

| Page                  | Status | Description                                                           |
|-----------------------|--------|-----------------------------------------------------------------------|
| Commercial Landing    | Live   | Public-facing page with live metrics refreshing every 60 seconds      |
| Institutional Page    | Live   | Track Record, PQC overview, competitor comparison, verifiable metrics |
| Credit/Lending Demo   | Live   | 8-checkpoint governance applied to credit risk evaluation             |
| Insurance Demo        | Live   | 8-checkpoint underwriting governance with risk assessment             |
| Energy Trading Demo   | Live   | 8-checkpoint energy trading governance (gas, oil, solar, wind, LNG)   |
| Biotech/Clinical Demo | Live   | 8-checkpoint clinical trial governance (real ClinicalTrials.gov data) |

All pages use the useLiveMetrics hook which refreshes data every 60 seconds from the Railway production API. Fallback values are used only when Railway is unreachable. System uptime days are calculated dynamically from Track Record start date (January 15, 2026).

## 21. Technical Validation — Terra/LUNA Forensic Simulation (May 2022)

The Terra/LUNA collapse of May 2022 destroyed over \$40 billion in market capitalization in 72 hours. No automated governance system in the market detected the structural failure before it became irreversible. OMNIX reconstructed this event using real historical market data to demonstrate how the 8-checkpoint fail-closed governance pipeline would have responded.

The simulation applied three key checkpoints — CP-0 (Signal Integrity Validator), CP-4 (Coherence Engine), and CP-7 (Temporal Coherence Validation) — at three critical timestamps before the collapse. The results demonstrate that OMNIX would have issued a WARNING at T-72h and a BLOCKED decision at both T-24h and T-6h, preserving 100% of position capital before the irreversible unwinding began.

This represents the first concrete demonstration of what OMNIX calls Architectural Certainty: a governance standard where the execution boundary is owned by the runtime, not orbited by it. The full forensic report is available as a separate document: OMNIX\_LUNA\_Forensic\_Simulation\_May2022.pdf (7 sections, 668 KB).

### 3-Phase Governance Results

| Phase            | Timestamp (UTC)     | LUNA Price | CP-0 SIV | CP-4 Coherence | CP-7 TCV | Decision              |
|------------------|---------------------|------------|----------|----------------|----------|-----------------------|
| Phase 1<br>T-72h | 2022-05-08<br>00:00 | \$68.84    | 88.9/100 | 77.7/100       | 56.8/100 | WARNING               |
| Phase 2<br>T-24h | 2022-05-10<br>00:00 | \$18.14    | 51.3/100 | 28.4/100       | 39.9/100 | BLOCKED               |
| Phase 3<br>T-6h  | 2022-05-10<br>18:00 | \$4.60     | 51.8/100 | 23.9/100       | 46.1/100 | BLOCKED<br>+ RECEIPT  |
| Collapse         | 2022-05-11<br>00:00 | \$1.73     | —        | —              | —        | ALL SYSTEMS<br>FAILED |

### Framework Comparison — Probabilistic vs. Forensic Governance

| Dimension          | Probabilistic Systems<br>(Industry Standard)    | OMNIX + VITT<br>(Forensic Governance)                    |
|--------------------|-------------------------------------------------|----------------------------------------------------------|
| Signal Validation  | Checks if data is statistically clean           | Forces signal to prove Logical Authenticity              |
| Confidence Model   | Inherits confidence from historical performance | Detects Manufactured Confidence; re-earned each cycle    |
| Regime Awareness   | Static thresholds, regime-agnostic              | HMM continuous estimation; thresholds adapt in real time |
| Temporal Coherence | Point-in-time validation only                   | Must be consistent with entire historical trajectory     |
| LUNA Outcome       | FAILED — did not detect Topological Collapse    | BLOCKED — Sovereign Gate activated at T-6h               |

## Cryptographic Governance Receipt (T-6h)

The T-6h BLOCKED decision generated a cryptographically signed governance receipt using Dilithium-3 (NIST-standardized post-quantum signature algorithm). The receipt records the exact checkpoint scores, failure reason, and regime classification. It is tamper-proof and publicly verifiable.

| Field           | Value                                                   |
|-----------------|---------------------------------------------------------|
| Decision        | <b>BLOCKED</b>                                          |
| Asset           | LUNA/USD                                                |
| Timestamp       | 2022-05-10T18:00:00+00:00                               |
| Price at Gate   | \$4.6044                                                |
| CP-0 SIV Score  | 51.76 / 100                                             |
| CP-4 Coherence  | 23.94 / 100                                             |
| CP-7 TCV Score  | 46.14 / 100                                             |
| Block Threshold | 65.0 / 100                                              |
| Regime          | CRASH                                                   |
| Failure Reason  | TEMPORAL_COHERENCE_VIOLATION + SIGNAL_INTEGRITY_FAILURE |
| SHA-256 Hash    | 3e2020dac7bc4e75265b454c98009ddd4fa87d73b4eef603        |
| Receipt Type    | FORENSIC_SIMULATION                                     |

## 22. Documents Generated

- Pitch deck for Eureka Dubai GCC 2026 (March 18 presentation)
- Business Model Canvas PDF
- Executive Report for investors
- Research Paper with reproducibility section
- Personalized mentor packages (multiple recipients)
- Q&A; Simulation — tough investor questions with prepared answers
- Presentation script with real operational data
- AI Risk Analyst Report (independent validation)
- 3-tier Security Documentation (public / institutional / internal)
- Internal Security Audit v1.0
- Validation questionnaires for industry professionals (multiple recipients)
- LinkedIn thought-leadership posts (English & Spanish)
- Terra/LUNA Forensic Simulation Report (OMNIX\_LUNA\_Forensic\_Simulation\_May2022.pdf)
- Technical Validation — Terra/LUNA for investor data room (OMNIX\_Technical\_Validation\_LUNA\_2022.pdf)

## 23. Summary in Numbers

| Metric                   | Value            | Notes                                       |
|--------------------------|------------------|---------------------------------------------|
| Total database tables    | 71               | PostgreSQL on Railway                       |
| Evaluation cycles        | 750,100          | shadow_trade_events rows                    |
| PQC-signed receipts      | 66,220           | Dilithium-3 signed                          |
| Vetos logged             | 25,681           | Real-time veto tracking                     |
| Real Kraken trades       | 2,237            | Phase 0 (Jul-Aug 2025)                      |
| Paper trades             | 119              | Learning Baseline (Nov-Jan)                 |
| Exit governance receipts | 78               | 3-gate exit pipeline                        |
| B2B clients registered   | 5                | With API key authentication                 |
| Sandbox sessions         | 3                | Test sessions created                       |
| Sandbox evaluations      | 6                | Evaluations run in sandbox                  |
| Vertical demos live      | 4                | Trading, Credit, Insurance, Energy, Biotech |
| Governance checkpoints   | 8                | CP-0 through CP-7 (fail-closed)             |
| Compliance modules       | 5                | NIST AI RMF / ISO 42001 / EU AI Act         |
| Development months       | ~3               | Dec 25, 2025 - Mar 13, 2026                 |
| Target event             | Eureka Dubai GCC | March 18, 2026 — Semifinalist               |

---

Report generated: March 14, 2026  
OMNIX Decision Governance Infrastructure — Abu Dhabi, UAE  
Internal document. Not externally audited. Not for distribution.